



INSTRUCTOR SOLUTION

Habib University
Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026
Instructor – Dr. Aamir Hasan - Section L4

Time = 25 Minutes

Quiz No 10

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly.

Most of the total score for the questions will be awarded on your reasoning.

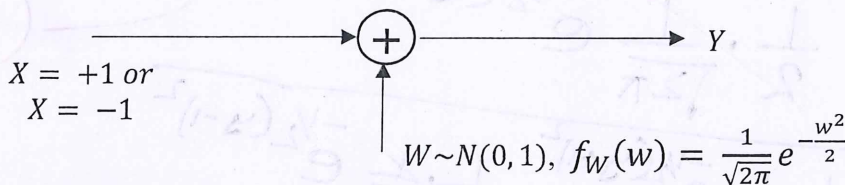
CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [20 points] [Bayes Rule - Inference]

Random Variable X is discrete and Y is continuous. The question pertains to the case (as discussed in the class) where we can observe Y and would like to infer X using the conditional probability, $P_{X|Y}(x|y)$. Assume that we know the prior distribution of X , $P_X(x)$ and the model $f_{Y|X}(y|x)$.

Part a [06 points] Derive the expression $P_{X|Y}(x|y)$. All necessary steps are to be shown.

Now assume the following system (see below) pertaining to the above-mentioned case that involves the discrete Random Variable X and a continuous Random Variable, Y . This is a simple model for a communication system.



Part b [07 points] Given $P(X = +1) = P(X = -1) = 0.5$, find and plot $P_{X|Y}(+1|y)$. Interpret the plot with regards to estimating X .

Part c [07 points] Now assume that $P(X = +1) = 0.75$ and $P(X = -1) = 0.25$. Find the revised $P_{X|Y}(+1|y)$ and plot it. Interpret the plot with regards to estimating X .

Note – provide necessary justifications.

$$(a) P(X=x, y \leq Y \leq y+\Delta) = P(X=x) \cdot P(y \leq Y \leq y+\Delta | X=x) \quad \text{--- 1.5}$$

$$\approx P_X(x) \cdot f_{Y|X}(y|x) \cdot \Delta \quad \text{--- Eq 1}$$

also,

$$P(X=x, y \leq Y \leq y+\Delta) = P(X=x | y \leq Y \leq y+\Delta) \cdot P(y \leq Y \leq y+\Delta) \quad \text{--- 1.5}$$

$$\approx P_{X|Y}(x|y) \cdot f_Y(y) \cdot \Delta \quad \text{--- Eq 2}$$

Equating (Eq 1) & (Eq 2)

$$P_{X|Y}(x|y) = P_X(x) \cdot f_{Y|X}(y|x) \quad \text{--- Infer x by observing y}$$

where $f_Y(y) = \sum_x P_X(x) \cdot f_{Y|X}(y|x)$ (2) --- Eq 3

(b) Using $(Eq 3) \Rightarrow P_{X|Y}(+1|y) = \frac{P_X(+1) \cdot f_{Y|X}(y|x=+1)}{f_Y(y)}$ (1)

What we know,

$P_X(+1) = 1/2$ (a)

$f_{Y|X}(y|+1) = \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2}$ (b) (Eq 4) (1)

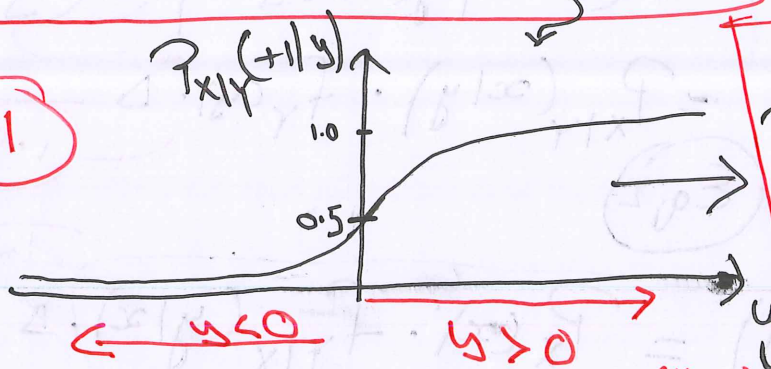
(c) $f_Y(y) = 1/2 \cdot f_{Y|X}(y|-1) + 1/2 \cdot f_{Y|X}(y|+1) \rightarrow TPT$

$= \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y+1)^2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2}$ (1) (Eq 5)

$P_{X|Y}(+1|y) = \frac{\frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2}}{\frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y+1)^2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2}}$ (1)

$= \frac{e^{-1/2(y-1)^2}}{e^{-1/2(y+1)^2} + e^{-1/2(y-1)^2}} = e^{-1/2(y-1)^2} \cdot \frac{1}{e^{-1/2(y-1)^2} \left(1 + \frac{e^{-1/2(y+1)^2}}{e^{-1/2(y-1)^2}} \right)}$

$P_{X|Y}(+1|y) = \frac{1}{1 + e^{-2y}}$ (Eq 6) (1)



The plot is of Probabilities for $x=+1$ Conditioned on differed values of y

The plot says that whenever the observation is greater than 0, then $P_{X|Y}(+1|Y>0) \geq 0.5$. So $X=+1$ is more likely than $X=-1$ (threshold).

(c) $P_X(+1) = 0.75$ and $P_X(-1) = 0.25$

Again using (Eq 3)

$$P_{X|Y}(+1|y) = \frac{P_X(+1) \cdot f_{Y|X}(y|+1)}{f_Y(y)} \quad \text{--- (Eq 7)}$$

$$P_X(+1) = 3/4$$

$$f_{Y|X}(y|+1) = \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2}$$

$$\begin{aligned} f_Y(y) &= \frac{1}{4} \cdot f_{Y|X}(y|-1) + \frac{3}{4} \cdot f_{Y|X}(y|+1) \quad \text{--- (1)} \\ &= \frac{1}{4} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y+1)^2} + \frac{3}{4} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2(y-1)^2} \end{aligned}$$

$$P_{X|Y}(+1|y) = \frac{\cancel{3/4} \cdot \cancel{1/\sqrt{2\pi}} e^{-1/2(y-1)^2}}{\cancel{1/4} \cdot \cancel{1/\sqrt{2\pi}} e^{-1/2(y+1)^2} + \cancel{3/4} \cdot \cancel{1/\sqrt{2\pi}} e^{-1/2(y-1)^2}} \quad \text{--- (Eq 8)}$$

Solving this

$$P_{X|Y}(+1|y) = \frac{1}{1 + 1/3 e^{-2y}} = \frac{3}{3 + e^{-2y}} \quad \text{--- (Eq 9)}$$

Let's determine y where $P_{X|Y}(+1|y) = 0.5$

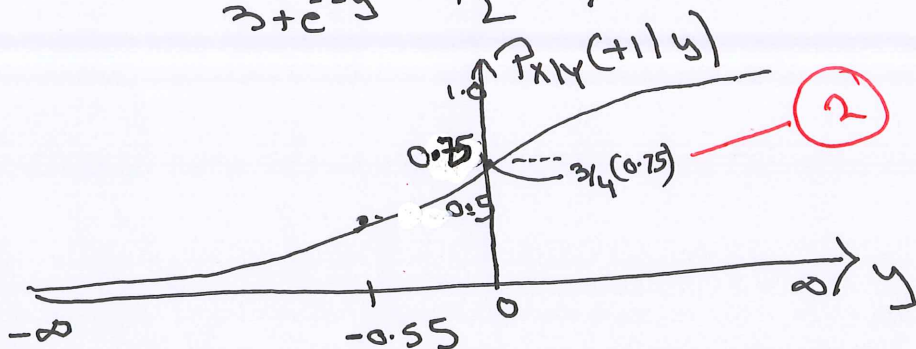
$$\frac{3}{3 + e^{-2y}} = \frac{1}{2} \Rightarrow 3 + e^{-2y} = 6 \Rightarrow e^{-2y} = 3$$

$$-2y = \ln 3$$

$$y = -\frac{1}{2} \ln 3$$

$$y = -0.55$$

reversed threshold



when $y=0$
 $P_{X|Y}(+1|0) = 0.75$

when $y \geq 0.55$, $P_{X|Y}(+1|y) \geq 0.5$ and when $y \leq -0.55$, $P_X(+1)$ is more likely than $P_X(-1)$ --- (1)